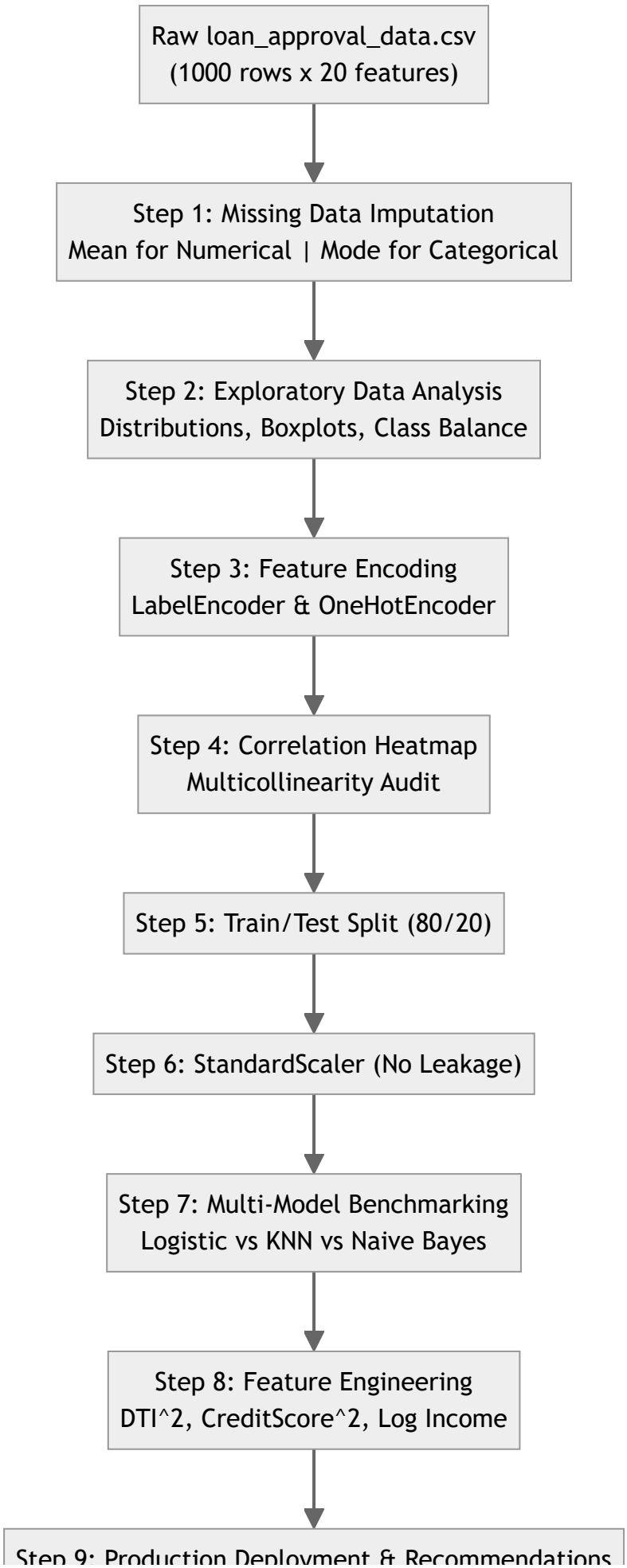
Lecture 27.02: Project Architecture & Dataset Ingestion

📌 Topic Overview

In this lecture, Shraddha ma'am outlines the **End-to-End Project Pipeline Architecture** and inspects the raw `loan_approval_data.csv` dataset containing 1,000 applicant records.

🏛️ End-to-End Pipeline Blueprint


Raw loan_approval_data.csv
(1000 rows x 20 features)
Step 1: Missing Data Imputation
Mean for Numerical | Mode for Categorical
Step 2: Exploratory Data Analysis
Distributions, Boxplots, Class Balance
Step 3: Feature Encoding
LabelEncoder & OneHotEncoder
Step 4: Correlation Heatmap
Multicollinearity Audit
Step 5: Train/Test Split (80/20)
Step 6: StandardScaler (No Leakage)
Step 7: Multi-Model Benchmarking
Logistic vs KNN vs Naive Bayes
Step 8: Feature Engineering
DTI^2, CreditScore^2, Log Income
Step 9: Production Deployment & Recommendations

Step 9: Production Deployment & Recommendations

📋 Feature Taxonomy (Classroom Table)

The dataset comprises 20 attributes:

- **Identifier**: `Applicant_ID` (drop before modeling).
- **Financial Features**: `Applicant_Income`, `Coapplicant_Income`, `DTI_Ratio`, `Savings`, `Collateral_Value`, `Loan_Amount`, `Loan_Term`.
- **Credit Bureau**: `Credit_Score`, `Existing_Loans`.
- **Demographics**: `Age`, `Employment_Status`, `Marital_Status`, `Dependents`, `Education_Level`, `Property_Area`.
- **Target Label**: `Loan_Approved` ($1 = \text{Approved}, 0 = \text{Rejected}$).

💻 Python Implementation

```
import pandas as pd

def inspect_loan_dataset_schema(csv_path="loan_approval_data.csv"):
    """
    Loads and audits the CreditWise loan dataset schema:
    1000 applicants across 20 attributes (demographics, income, credit bureau, tar
    """
    df = pd.read_csv(csv_path)

    summary = {
        'shape': df.shape,
        'features': list(df.columns),
        'missing_counts': df.isnull().sum()[df.isnull().sum() > 0].to_dict(),
        'target_distribution': df['Loan_Approved'].value_counts(normalize=True).to
    }
    return summary

# --- Function Call & Testing ---
# report = inspect_loan_dataset_schema()
# print("Dataset Dimensions:", report['shape'])
# print("Features with Missing Values:", report['missing_counts'])
# print("Target Class Proportions:", report['target_distribution'])
```

🔑 Key Takeaways

1. A structured ML project follows a strict sequential pipeline.
2. Identifiers (`Applicant_ID`) must be dropped to prevent data leakage.